Question 2: LONGEST ARITHMETIC SUBSEQUENCE

Aim:
To find the length of the longest arithmetic subsequence in an array using Dynamic Programming.

Algorithm:
1. Create a DP table.
2. For each pair of elements, calculate their difference.
3. Store subsequence lengths based on differences.
4. Update the maximum length found.
5. Return the maximum length.

Pseudocode:
```
Input n, arr[]
Create dp array of dictionaries
ans = 1
For i from 0 to n-1:
    For j from 0 to i-1:
        diff = arr[i] - arr[j]
        dp[i][diff] = dp[j].get(diff,1) + 1
        ans = max(ans, dp[i][diff])
Return ans
```

Python Program:
```python
def longest_arithmetic_subsequence(arr):

    n = len(arr)

    if n <= 1:

        return n

    dp = [dict() for _ in range(n)]

    ans = 2

    for i in range(n):

        for j in range(i):

            diff = arr[i] - arr[j]
```

```
            dp[i][diff] = dp[j].get(diff, 1) + 1

            ans = max(ans, dp[i][diff])

    return ans

arr = [3, 6, 9, 12, 15]

print("Length =", longest_arithmetic_subsequence(arr))
```

Sample Input:
n = 5
arr = [3, 6, 9, 12, 15]

Sample Output:
Length = 5

Time Complexity: $O(n^2)$

Space Complexity: $O(n^2)$

Result:
The length of the longest arithmetic subsequence was successfully determined using Dynamic Programming.